C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 1: Even or Odd

Write a program to check whether a number entered by the user is even or odd.

Logic: If n % 2 == 0 → Even else Odd.

Explanation (Hindi): □□ □□□□□□□□□□ □□□ □□□□ □□ □□ □□□□ □□□□ □□□□ '%' □□□□□□ □□ remainder □□□ □□□□; rem
=> even, □□□□ □□ odd□

```
Enter a number: 7 7 is an odd number.
```

Task 2: Sum/Diff/Product/Quotient

Input two integers and display sum, difference, product and quotient.

Logic: Use +, -, *, /. Watch integer division.

Explanation (Hindi): □□ □□□□ □□□□ +,-,* □□□□; division integer □□ □□ decimal □□□□□; float cast □□□□ □□□ decimal □□□□□□

```
Input: 8 3 Output: Sum=11 Diff=5 Prod=24
Quot=2
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 3: Largest of Three

Find and print the largest of three numbers.

Logic: Compare pairwise using if-else.

Explanation (Hindi): □□□ □□□□ □□□; max=a □□□□□ b □□ c □□ compare □□ □□ update □□□□□

```
Input: 5 9 2 Output: Largest = 9
```

Task 4: Positive/Negative/Zero

Check whether integer is positive, negative or zero.

Logic: Compare with 0: >0, <0, ==0.

Explanation (Hindi): □□□□ □□ 0 □□ compare □□□□ □□ result □□□□□□

```
Input: -4 Output: Negative
```

Task 5: Bitwise Even/Odd

Check even/odd using bitwise operator.

Logic: n & 1 == 0 → even.

Explanation (Hindi): LSB □□□□□; fast method□

```
Input: 10 Output: Even
```

Task 6: Leap Year

Determine whether a year is leap year.

Logic: (y%4==0 && y%100!=0) || y%400==0

Explanation (Hindi): Leap-year rules □□□□ □□□□□

```
Input: 2000 Output: Leap Year
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 7: Vowel or Consonant

Check whether an entered character is vowel or consonant.

Logic: Check char in set vowels.

Explanation (Hindi): Letter ko set me dekho; vowel nahi to consonant assume karo.

```
Input: a Output: Vowel
```

Task 8: Divisible by 3 and 5

Check whether number divisible by both 3 and 5.

Logic: n%3==0 && n%5==0

Explanation (Hindi): □□□□□ conditions true □□□ □□ yes□

```
Input: 15 Output: Yes
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

By: Ahmed Raza   Page: 5   Generated: 2025-10-26

Task 9: Swap Without Temp

Swap two numbers without third variable.

Logic: Use arithmetic or XOR trick.

Explanation (Hindi): Arithmetic swap □□□ overflow risk; XOR safe.

```
Input: 4 7 Output: After swap: 7 4
```

Task 10: Quadratic Roots

Find roots using discriminant.

Logic: D=b*b-4ac; check sign.

Explanation (Hindi): D □□ roots □□ □□□□□□□ □□ □□□□ □□□□□□□□□

```
Input: 1 -3 2 Output: Roots: 2 and 1
```

Task 11: Print 1 to N

Print numbers 1..N using for loop.

Logic: for(i=1;i<=N;i++) cout<<i;

Explanation (Hindi): Simple for loop.

```
Input:5 Output:1 2 3 4 5
```

Task 12: Print N to 1

Print numbers descending using while.

Logic: while(n>=1){ cout<<n; n--; }

Explanation (Hindi): While loop with decrement

```
Input:5 Output:5 4 3 2 1
```

Task 13: Sum of N Naturals

Compute sum of 1..N.

Logic: Use n*(n+1)/2.

Explanation (Hindi): Formula efficient.

```
Input:10 Output:55
```

Task 14: Sum of Evens

Sum even numbers up to N.

Logic: Loop step 2 or formula.

Explanation (Hindi): Even numbers summation।

```
Input:6 Output:12
```

Task 15: Sum of Odds

Sum odd numbers up to N.

Logic: Count k and use k^2.

Explanation (Hindi): First k=(N+1)/2 then k^2.

```
Input:5 Output:9
```

Task 16: Multiplication Table

Table of N from 1..10.

Logic: for i=1..10 print N*i.

Explanation (Hindi): Formatted output.

```
Input:3 Output:3x1=3...3x10=30
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 17: Factorial

Compute factorial iteratively.

Logic: fact=1; for i=1..n fact*=i;

Explanation (Hindi): Multiply 1..n.

```
Input:5 Output:120
```

Task 18: Reverse Digits

Reverse integer digits.

Logic: rev=rev*10 + n%10; n/=10;

Explanation (Hindi): Digit extraction and build reverse.

```
Input:123 Output:321
```

Task 19: Sum of Digits

Sum digits using % and /.

Logic: sum+=n%10; n/=10;

Explanation (Hindi): Add each digit.

```
Input:123 Output:6
```

Task 20: Count Digits

Count digits by dividing by 10.

Logic: count++ until n==0.

Explanation (Hindi): or use log10 for positives.

```
Input:7894 Output:4
```

Task 21: Palindrome Number

Reverse and compare.

Logic: If reverse==original then palindrome.

Explanation (Hindi): Reverse build and compare.

```
Input:121 Output:Palindrome
```

Task 22: Armstrong (3-digit)

Sum cubes of digits equals number.

Logic: Sum d^3 and compare.

Explanation (Hindi): For 153 works: 1^3+5^3+3^3=153.

```
Input:153 Output:Armstrong
```

Task 23: Perfect Number

Sum proper divisors equal number.

Logic: Sum divisors and compare.

Explanation (Hindi): Use sqrt optimization for divisors.

```
Input:28 Output:Perfect Number
```

Task 24: Power without pow

Compute power by loop.

Logic: res=1; for i in range exp multiply.

Explanation (Hindi): Or use fast exponentiation for speed.

```
Input:2 5 Output:32
```

Task 25: Fibonacci N terms

Generate Fibonacci sequence.

Logic: a=0,b=1; next=a+b; update.

Explanation (Hindi): Iterative generation.

```
Input:6 Output:0 1 1 2 3 5
```

Task 26: Prime Check

Check up to sqrt(n) for divisors.

Logic: for i=2..sqrt(n) if divides then not prime.

Explanation (Hindi): Optimization: skip even numbers after 2.

```
Input:29 Output:Prime
```

Task 27: GCD Euclid

Euclidean GCD algorithm.

Logic: while(b){ r=a%b; a=b; b=r; }

Explanation (Hindi): Efficient GCD method.

```
Input:12 15 Output:GCD=3
```

Task 28: LCM via GCD

LCM = (a/gcd)*b

Logic: Compute gcd first.

Explanation (Hindi): Avoid large intermediate multiply.

```
Input:4 6 Output:LCM=12
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 29: Primes up to N

List primes using check or sieve.

Logic: Check each number's primality.

Explanation (Hindi): Sieve faster for large N.

```
Input:10 Output:2 3 5 7
```

Task 30: Sum Primes up to N

Sum primes found up to N.

Logic: Accumulate primes.

Explanation (Hindi): Use sieve for big N.

```
Input:10 Output:17
```

Task 31: Star Triangle

Print triangle of stars.

Logic: Nested loops: rows & stars.

Explanation (Hindi): Inner loop prints stars equal to row.

```
Input:3 Output:* \n* * \n* * *
```

Task 32: Inverted Star

Inverted triangle.

Logic: Nested loops decreasing.

Explanation (Hindi): Start from N to 1.

```
Input:3 Output:* * * \n* * \n*
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 33: Number Triangle

Row i contains 1..i

Logic: Nested loops printing numbers

Explanation (Hindi): Basic nested loops

```
Input:4 Output:1 \n1 2 \n1 2 3 \n1 2 3 4
```

Task 34: Pattern 1 22 333

Print i repeated i times each row

Logic: Inner loop prints i times

Explanation (Hindi): Repetition logic per row

```
Input:4 Output:1 \n22 \n333 \n4444
```

Task 35: Alphabet Triangle

Triangle of consecutive letters

Logic: Use char starting from 'A' and increment

Explanation (Hindi): Increment char per print

```
Input:3 Output:A \nB C \nD E F
```

Task 36: Centered Pyramid

Centered pyramid with spaces and stars

Logic: For row i print (N-i) spaces and 2*i-1 stars

Explanation (Hindi): Spacing + odd stars

```
Input:3 Output:centered pyramid
```

Task 37: Hollow Square

Square with hollow inside

Logic: Print star for border else space

Explanation (Hindi): Border condition check

```
Input:4 Output:hollow square
```

Task 38: Hollow Triangle

Hollow right-angled triangle

Logic: Print stars on borders and base

Explanation (Hindi): Use j==1 || j==i || i==N

```
Input:5 Output:hollow triangle
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 39: Palindromic Pyramid

Print palindromic numbers per row

Logic: Print ascending then descending numbers

Explanation (Hindi): Two inner loops for symmetry

```
Input:4 Output:1 \n121 \n12321 \n1234321
```

Task 40: Floyd's Triangle

Increasing numbers row-wise

Logic: Keep counter k and increment

Explanation (Hindi): Global counter across rows

```
Input:4 Output:1 \n2 3 \n4 5 6 \n7 8 9
10
```

Task 41: Strong Number

Sum factorial of digits equals number

Logic: Compute factorial per digit and sum

Explanation (Hindi): Compare sum with original

```
Input:145 Output:Strong Number
```

Task 42: Armstrong List

List Armstrong numbers 1..1000

Logic: Check sum of digit powers equals number

Explanation (Hindi): Loop and test per number

```
Output:1 153 370 371 407
```

Task 43: Sum Squares

Sum squares 1..N

Logic: Use formula n(n+1)(2n+1)/6

Explanation (Hindi): Efficient formula

```
Input:3 Output:14
```

Task 44: Sum Cubes

Sum cubes 1..N

Logic: Use (n(n+1)/2)^2 formula

Explanation (Hindi): Square of sum

```
Input:3 Output:36
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 45: Binary to Decimal

Convert binary to decimal

Logic: dec=dec*2 + lastBit

Explanation (Hindi): Iterate bits and accumulate

```
Input:1011 Output:11
```

Task 46: Decimal to Binary

Convert decimal to binary

Logic: Collect remainders n%2 and reverse

Explanation (Hindi): Reverse remainders to get bits

```
Input:6 Output:110
```

Task 47: Find Factors

Print factors of n

Logic: Check i up to sqrt(n) and pair i and n/i

Explanation (Hindi): Use sqrt optimization

```
Input:28 Output:1 2 4 7 14 28
```

Task 48: Perfect Square

Check perfect square

Logic: r=int(sqrt(n)); if r*r==n yes

Explanation (Hindi): sqrt and verify

```
Input:49 Output:Perfect square
```

Task 49: Tables 1..10

Print multiplication tables 1..10

Logic: Nested loops i=1..10 j=1..10

Explanation (Hindi): Two nested loops produce table

```
Partial Output:1x1=1 ...
```

Task 50: Min Max Digit

Smallest and largest digit in number

Logic: Iterate digits and track min/max

Explanation (Hindi): Init min=9 max=0 update per digit

```
Input:38402 Output:Min=0 Max=8
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

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Task 51: Buzz Number

Divisible by 7 or ends with 7

Logic: n%7==0 || n%10==7

Explanation (Hindi): OR condition for buzz

```
Input:17 Output:Buzz
```

Task 52: Duck Number

Contains zero but not starting with zero

Logic: String check for '0' and s[0]!='0'

Explanation (Hindi): Convert to string and check

```
Input:105 Output:Duck
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 53: Spy Number

Sum digits == product digits

Logic: Compute both and compare

Explanation (Hindi): Calculate sum and product and compare

```
Input:1124 Output:Spy
```

Task 54: Trimorphic

n^3 ends with n

Logic: Check (n^3) % (10^d) == n

Explanation (Hindi): Modulo with 10^digits

```
Input:4 Output:Trimorphic
```

Task 55: List Spy

List spy numbers 1..1000

Logic: Check spy property per number

Explanation (Hindi): Loop and print matching

```
Output:(partial)
```

Task 56: Kaprekar

Kaprekar by splitting square

Logic: Square n and split digits to two parts

Explanation (Hindi): Try splits and check sums

```
Input:45 Output:Kaprekar
```

Task 57: List Kaprekar

List Kaprekar numbers up to 1000

Logic: Test kaprekar for each

Explanation (Hindi): Loop and print matches

```
Output:1 9 45 55 99 297
```

Task 58: Magic Number

Digital root equals 1

Logic: Repeat sumDigits until single digit, check ==1

Explanation (Hindi): Repeat process until single digit

```
Input:1729 Output:(depends)
```

Task 59: Next Palindrome

Find smallest palindrome > N

Logic: Increment and check palindrome

Explanation (Hindi): Bruteforce works for small N

```
Input:123 Output:131
```

Task 60: Palindromes Interval

List palindromes between A and B

Logic: Check each number reverse==original

Explanation (Hindi): Range loop and test

```
Input:100 150 Output:101 111 121 131 141
```

Task 61: Fibonacci <=1000

Generate fib numbers <= 1000

Logic: Iterative until next>1000

Explanation (Hindi): Stop when next exceeds limit

```
Output:0 1 1 2 3 5 ... 987
```

Task 62: Twin Prime

n and n+2 both prime

Logic: isPrime(n) && isPrime(n+2)

Explanation (Hindi): Check both numbers for primality

```
Input:11 Output:Twin Prime
```

Task 63: List Twin Primes

List twin primes up to 100

Logic: Check k and k+2 primality

Explanation (Hindi): Print matching pairs

```
Output:(3,5)(5,7)(11,13)
```

Task 64: Happy Number

Sum squares digits repeat until 1

Logic: If reaches 1 happy else cycle

Explanation (Hindi): Detect cycle using set or known 4 loop

```
Input:19 Output:Happy
```

Task 65: Sum First N Primes

Sum first N primes

Logic: Find and add primes until count N

Explanation (Hindi): Maintain count and accumulate

```
Input:5 Output:28
```

Task 66: Butterfly Pattern

Print butterfly pattern

Logic: Nested loops left-right with spaces

Explanation (Hindi): Mirror top and bottom wings

```
Input:4 Output:butterfly pattern
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 67: Pascal w/o arrays

Pascal values using combinatorics

Logic: val = val*(row-k+1)/k iterative

Explanation (Hindi): Compute next from previous

```
Output:Pascal rows
```

Task 68: Inc-Dec Pattern

Increase then decrease number rows

Logic: First ascending then descending

Explanation (Hindi): Two phase nested loops

```
Input:4 Output:pattern
```

Task 69: Alt 1/0 Pattern

Alternating 1 and 0

Logic: Use (i+j)%2 parity

Explanation (Hindi): Parity decides bit value

```
Input:4 Output:pattern
```

Task 70: Numeric Diamond

Diamond with continuous numbers

Logic: Use counter and mirror rows

Explanation (Hindi): Increment counter and mirror

```
Input:4 Output:numeric diamond
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

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Task 71: Sum Factorial Digits

Sum factorials of digits compare to n

Logic: sum += fact(d) per digit

Explanation (Hindi): Compute factorial per digit and sum

```
Input:145 Output:Strong
```

Task 72: Neon Number

Sum digits of square equals n

Logic: s=n*n; sumDigits(s)==n

Explanation (Hindi): Square and sum its digits

```
Input:9 Output:Neon
```

Task 73: Automorphic

n^2 ends with n

Logic: (n*n) % (10^d) == n

Explanation (Hindi): Modulo check with 10^digits

```
Input:25 Output:Automorphic
```

Task 74: Digital Root

Repeated sum until single digit

Logic: Use 1+(n-1)%9 formula for n>0

Explanation (Hindi): Fast formula available

```
Input:9875 Output:2
```

Task 75: Harshad

n divisible by sum of digits

Logic: n % sumDigits(n) == 0

Explanation (Hindi): Compute digit sum then check divisibility

```
Input:18 Output:Harshad
```

Task 76: Product of Digits

Product of digits

Logic: prod*=d while extracting digits

Explanation (Hindi): Zero digit makes product zero

```
Input:1234 Output:24
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 77: Count Prime Digits

Count digits in {2,3,5,7}

Logic: Check each digit set membership

Explanation (Hindi): Increment counter for matches

```
Input:2394 Output:3
```

Task 78: First N Primes

Print first N primes

Logic: Find primes sequentially until count N

Explanation (Hindi): Candidate testing and count

```
Input:5 Output:2 3 5 7 11
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 79: Triangular Series

Print triangular numbers t_n

Logic: Use formula n(n+1)/2

Explanation (Hindi): Loop and print formula results

```
Input:5 Output:1 3 6 10 15
```

Task 80: Alternating Sign

1,-2,3,-4...

Logic: Use sign based on parity

Explanation (Hindi): Odd positive even negative

```
Input:6 Output:1 -2 3 -4 5 -6
```

Task 81: Sum Factorial Series

Sum 1!+2!+...+N!

Logic: Compute factorial iteratively and add

Explanation (Hindi): running factorial saves recompute

```
Input:4 Output:33
```

Task 82: Pascal Combinatorics

Print Pascal triangle rows

Logic: Use multiplicative nCk update

Explanation (Hindi): Compute values iteratively

```
Output:Pascal rows
```

Task 83: Odds 100..200

Print odd numbers between 100 and 200

Logic: Start at 101 step by 2

Explanation (Hindi): Loop stepping gives odd numbers

```
Output:101 103 ... 199
```

Task 84: Div7 not5 upto N

Divisible by 7 but not 5

Logic: Check n%7==0 && n%5!=0

Explanation (Hindi): Filter with both conditions

```
Input:50 Output:7 14 21 28 42 49
```

Task 85: Even Digit-Sum

Numbers with even digit-sum up to N

Logic: if sumDigits(i)%2==0 then print

Explanation (Hindi): Compute digit-sum and check parity

```
Partial Output
```

Task 86: Palindromes 1..1000

List palindromes

Logic: Check reverse==original

Explanation (Hindi): Loop and test each

```
Output:palindromes
```

Task 87: Hollow Diamond

Hollow diamond of stars

Logic: Spaces + edge star conditions

Explanation (Hindi): Print mirrored triangles with edges

```
Output:diamond
```

Task 88: Alphabet Right Triangle

Right-angled alphabet triangle

Logic: Increment char per print

Explanation (Hindi): Nested loops with char increment

```
Output:pattern
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 89: Inverted Alphabet

Inverted alphabet triangle

Logic: Decrease counts per row

Explanation (Hindi): Use nested loops decreasing

```
Output:pattern
```

Task 90: Palindrome Alphabet Pyramid

A, ABA, ABCBA pattern

Logic: Print ascending then descending letters

Explanation (Hindi): Combine inner loops for palindrome

```
Output:pattern
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 91: Automorphic List

List automorphic numbers 1..1000

Logic: Check modulo 10^d on square

Explanation (Hindi): Print matches like 5,6,25,76

```
Output:examples
```

Task 92: Kaprekar List

List Kaprekar numbers up to 1000

Logic: Split square and check sums

Explanation (Hindi): Print matches 1,9,45...

```
Output:1 9 45 55 99 297
```

Task 93: Perfect Cube

Check perfect cube using cbrt

Logic: r=round(cbrt(n)); if r^3==n then yes

Explanation (Hindi): Use cbrt and verify integer cube

```
Input:27 Output:Perfect cube
```

Task 94: Power of Two

Check power of two

Logic: n>0 && (n & (n-1))==0

Explanation (Hindi): Bitwise single set bit trick

```
Input:8 Output:Power of 2
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

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Task 95: Binary Without Arrays

Binary of decimal without arrays

Logic: Collect remainders and reverse

Explanation (Hindi): Store remainders then reverse order

```
Input:13 Output:1101
```

Task 96: Strong Numbers List

List strong numbers 1..1000

Logic: Sum factorial digits equals number

Explanation (Hindi): Print matches 1,2,145

```
Output:1 2 145
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 97: Count Vowels/Consonants/Digits

Classify chars in string

Logic: Use isalpha/isdigit and vowel set

Explanation (Hindi): Iterate chars and count types

```
Input:Ab2c Output:V=1 C=2 D=1
```

Task 98: Reverse String

Reverse string without library

Logic: for i=len-1..0 print s[i]

Explanation (Hindi): Loop from last index to first

```
Input:Hello Output:olleH
```

C++ 100 Practical Programs — Pro Notebook (Hindi)

Task 99: Squares Series

Print squares 1^2..N^2

Logic: for i=1..N print i*i

Explanation (Hindi): Compute by multiplication

```
Input:5 Output:1 4 9 16 25
```

Task 100: Numeric Diamond

Numeric diamond continuous numbers

Logic: Use counter and mirror rows

Explanation (Hindi): Careful counters and spacing for symmetry

```
Input:4 Output:numeric diamond
```